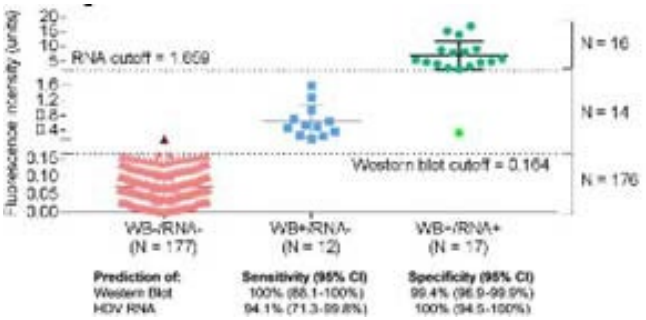
16/17 who were HDV RNA positive (sensitivity = 94.1%); specificity was 99.4% for anti-HDV and 100% for HDV RNA (Figure). The disease comparison included 431 cases (HCC, n = 312; cirrhosis only, n = 119), and 470 controls. Compared to controls, cases more often had HBV monoinfection (HBsAg+/HDV RNA-) (54.1% vs 17.0%; p < 0.001) and HBV-HDV coinfection (HBsAg+/HDV RNA+) (3.9% vs 0%; p < 0.001). After adjusting for covariates (including age, sex, socioeconomic status, alcohol, smoking, body mass index, anti-HCV, and *TP53* Ser-249), an increased risk of HCC or cirrhosis was observed for both HBV monoinfected (adjusted odds ratio [aOR] = 8.35; 95%CI = 5.80–12.03) and HBV-HDV coinfected (unadjusted OR = infinity; 95%CI = 9.49-infinity) compared to uninfected people. Risk estimates were similar when HDV status was classified by anti-HDV status.

Conclusion: HDV infection contributes to the high incidence of HCC and cirrhosis in The Gambia. HDV Q-MAC had high sensitivity and specificity for both anti-HDV and HDV RNA positivity in this population. This new assay could be useful for future studies of HDV infection in other African populations.



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Disclosure of Interest: P. Mahale: None Declared, P. Aka: None Declared, X. Chen: None Declared, R. Pfeiffer: None Declared, P. Liu: None Declared, S. Groover: None Declared, M. Mendy: None Declared, R. Njie: None Declared, J. Goedert: None Declared, G. Kirk: None Declared, J. Glenn Consultant for: Eiger BioPharmaceuticals, Inc., T. O'Brien: None Declared.

P1-135 | Molecular detection of Hepatitis B virus (HBV) DNA among Hepatitis B Surface antigen (HBsAg) Negative Blood Donors in Angola

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Hepatitis B virus (HBV) donor testing is a routine procedure in blood centers to guarantee adequate safe blood. Hepatitis B surface antigen (HBsAg), identified in early HBV infection, is the major serological marker used for screening HBV infection. Laboratory results obtained from the available commercial methods may vary in sensitivity and specificity to HBsAg according to serological assay used and can range in sensitivity from 0.013 to >4 IU/mL with the International Consortium for Blood Safety Quantitative Panel. Rapid tests for HBsAg detection carry a low analytical sensitivity of >1.5 IU/mL and are frequently used in locations that lack laboratory facilities. Furthermore, HBsAg-negative HBV infected individuals, that could be missed with serological screening may be detected through nucleic acid testing, nonetheless, is rarely performed in many HBV endemic regions where lifesaving transfusions are common. In this study we investigate the presence of HBV DNA in stored samples collected from blood donors in 2007 in a province located in the central part of Angola that tested negative for HBV and other transfusion-transmissible infections by immunochromatographic rapid testing. A total of 173 serum samples with volumes greater than 2 mL were selected and submitted to HBV DNA detection by a real time quantitative PCR assay. Samples with detectable HBV DNA, were retested for HBsAg by a different commercial rapid test device and a quantitative HBsAg assay. Ten samples (5.8%) had detectable HBV DNA, of which five had detectable HBsAg. Among those positive for HBsAg three were positive for both HBsAg assays and two solely on the quantitative HBsAg assay. The two samples positive exclusively on the quantitative HBsAg assay had low titers of HBsAg, both less than 0.09 IU/mL. The viral loads for HBV DNA varied from below the limit of detection to $\log_{10}$ 7.58. The absence of HBsAg in the presence of HBV DNA, calls our attention to occult HBV infection identified in 3% (5/173) of the blood donor samples studied. These results demonstrate the importance of sensitive HBsAg screening assays in blood centers along with HBV DNA testing to avoid further spread of HBV infection, especially in countries highly endemic for HBV.

Disclosure of Interest: None Declared.